

TITLE PAGE

- **Problem Statement ID** – 26172
- **Problem Statement Title-** Low Latency and Efficient Voice Activator for Edge Devices
- **Theme-** Smart Automation
- **PS Category-** Hardware
- **Team ID-**
- **Team Name -** EDITH

Problem:

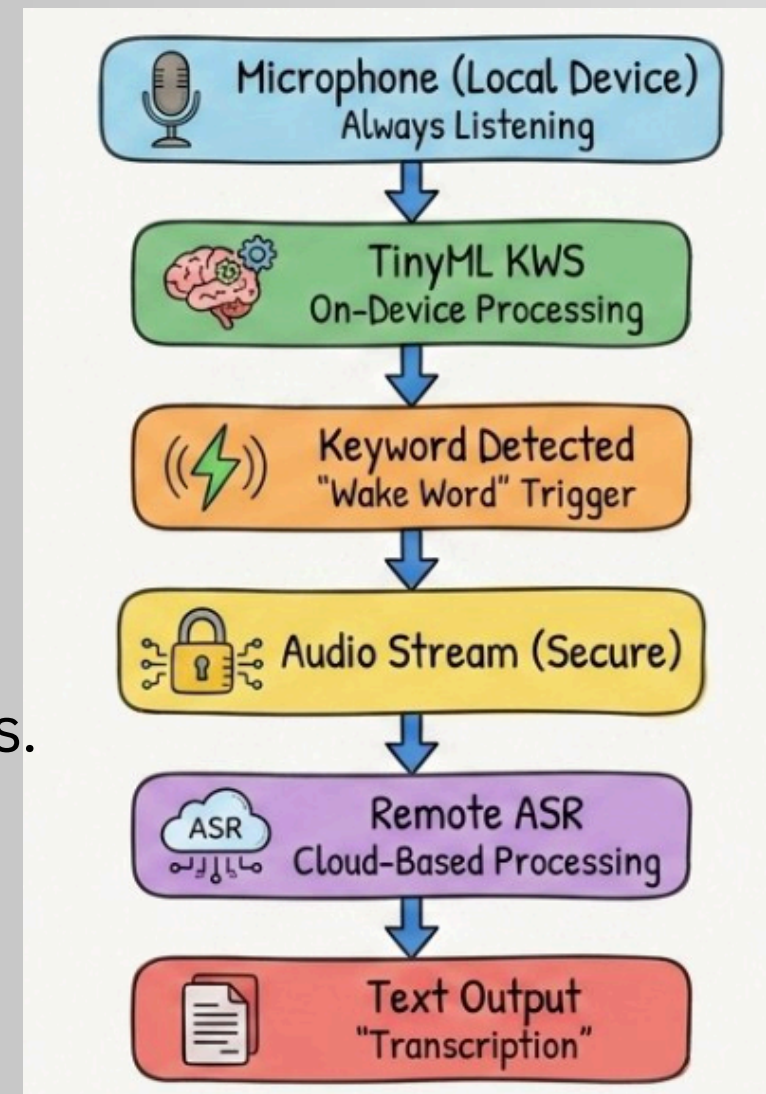
Cloud-based voice activation causes latency, bandwidth overhead, privacy concerns, and network dependency, making it unsuitable for resource-constrained critical environments.

Our Idea:

EDITH uses a hybrid Edge–Cloud architecture: a lightweight TinyML model detects a custom keyword locally, then streams only the following speech to a remote ASR server.

Proposed Solution:

- Custom **TinyML KWS** for local keyword detection.
- **<256 KB RAM & <10%** idle CPU optimized edge operation.
- Instant audio streaming after keyword detection.
- Ring buffer to prevent command audio loss.
- Robust testing under noise and multiple speakers.



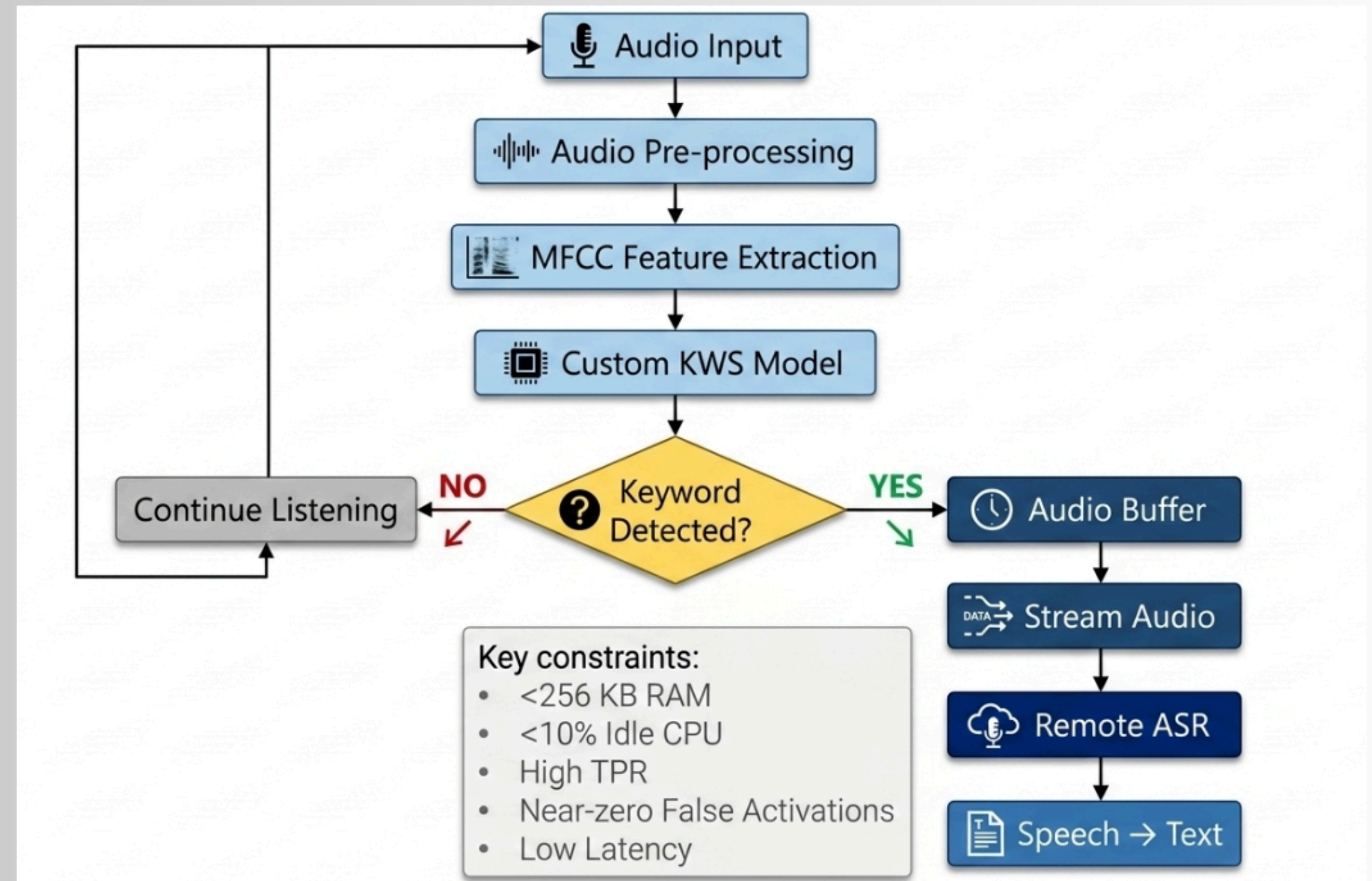
Innovation / Uniqueness:

- **Multilingual Keyword Support** – Custom keywords can be trained in any language, enabling wider and more inclusive voice activation.
- **Privacy-Aware Streaming** – Audio transmitted only after local keyword detection.
- **Multi-Speaker Robustness** – Handles different speakers, accents and noise.
- **Circular Audio Buffer** – Prevents loss of speech after keyword detection.

Technologies to be Used:

- **Hardware:** ESP32 / low-power MCU, MEMS microphone
- **Programming:** Python, C/C++
- **AI/ML:** TensorFlow Lite for Microcontrollers, TinyML
- **Audio Processing:** Log-Mel Spectrograms
- **Model Optimization:** INT8 quantization, memory & CPU optimization
- **Communication:** Lightweight audio streaming over Wi-Fi
- **Server:** Remote ASR server for speech-to-text processing

Methodology & Implementation:





Potential Challenges & Risks:

Challenge

Background noise

Multiple speakers

Limited RAM & CPU

Detection delay

Network delay

False activations

Risk

False activations / missed keywords

Reduced detection accuracy

Model may exceed hardware constraints

Beginning of command may be lost

Increased ASR latency

Unwanted audio transmission

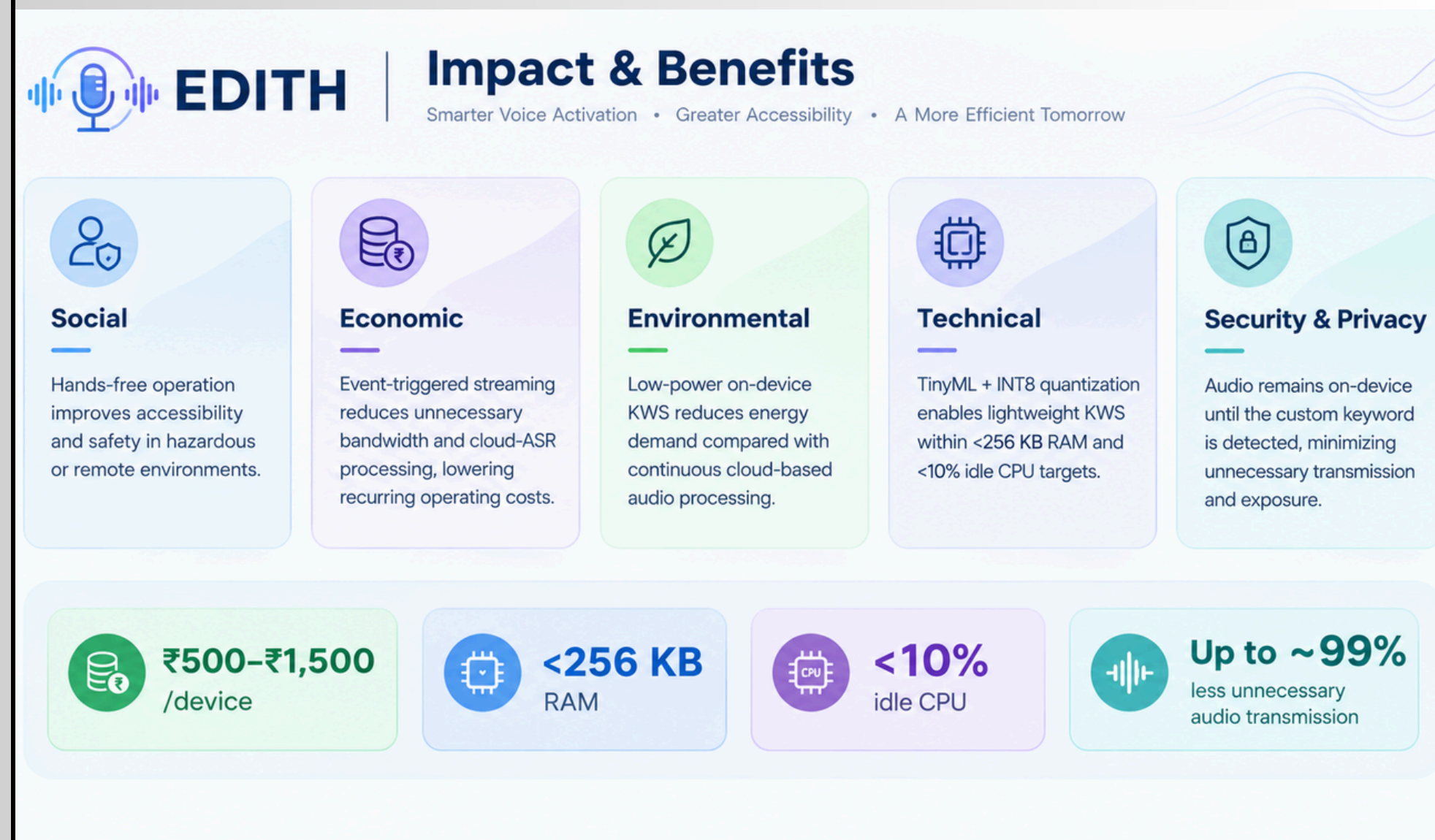
Strategies for Overcoming Challenges

- **Noise augmentation & diverse datasets** → Improve robustness against background noise and different speakers.
- **INT8 quantization & model optimization** → Reduce RAM, Flash and CPU usage.
- **Confidence thresholding & temporal smoothing** → Minimize false activations.
- **Circular audio buffer** → Prevent loss of the beginning of commands.
- **Efficient streaming & buffering** → Reduce ASR communication latency.
- **Extensive real-world testing** → Validate performance across speakers, noise levels and distances.

Potential Impact on Target Audience

- **Critical Operations:** Delivers reliable voice activation in space, defense, and industrial zones lacking stable connectivity.
- **Faster Response:** Cuts activation and network delays by **~30%** via instant, local edge processing.
- **Absolute Privacy:** Guarantees data security by keeping continuous audio **100% on-device** until triggered.
- **Field Accessibility:** Provides seamless, hands-free machine interaction in high-intensity, rugged workspaces.
- **Edge Reliability:** Functions flawlessly on minimal computing resources despite intermittent network outages.

Benefits of the solution (social, economic, environmental, etc.)



Details / Links of the Reference and Research Work

- “Hello Edge: Keyword Spotting on Microcontrollers” Yundong Zhang, Naveen Suda, Liangzhen Lai & Vikas Chandra, Arm, 2017 [link](#)
- “Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition” Pete Warden, Google Brain, 2018 [link](#)
- “TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems” Robert David et al., 2021 [link](#)